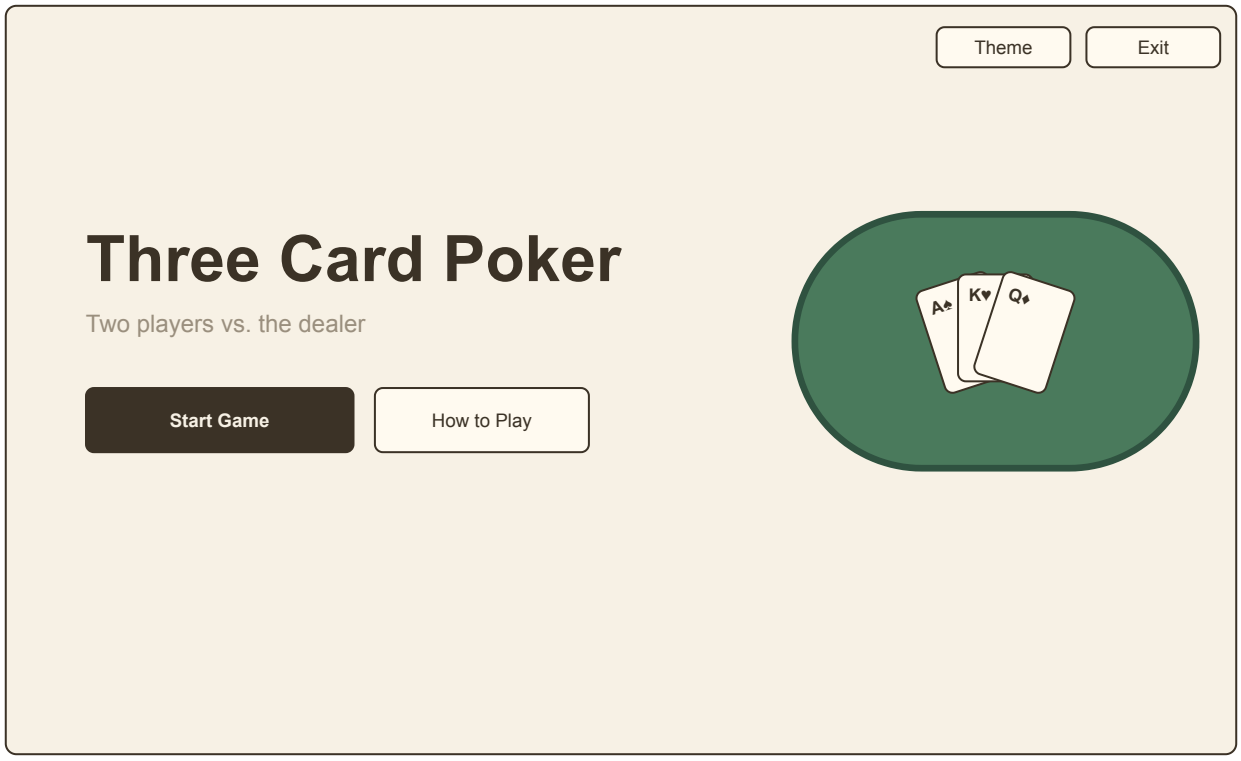
Three Card Poker — Part 1

Angelo Landiza — CS 342

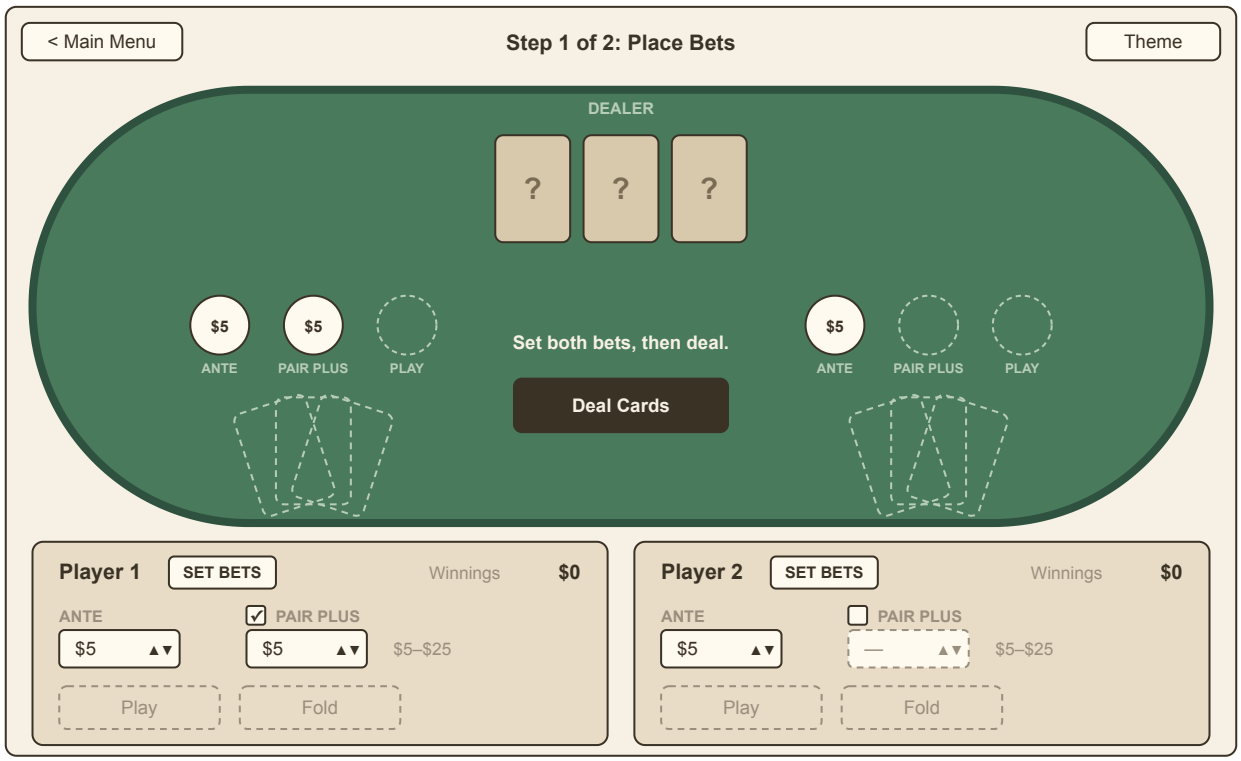
1. Wireframes

Screen 1 — Welcome / Main Menu



Transition: screens fade out / in (FadeTransition).

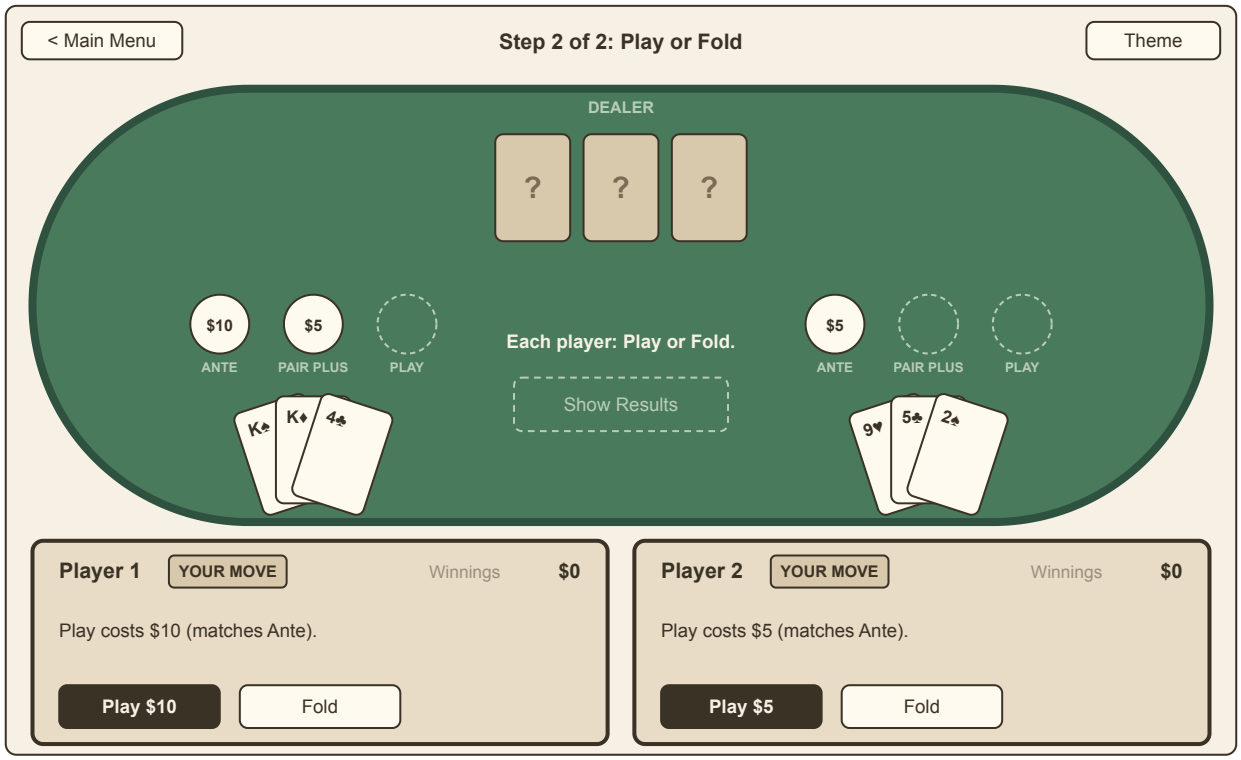
Screen 2 — Game Play (Step 1: Bet)



Spinners stop at $5 and $25.

Main Menu mid-round asks: "Leave game? Progress will be lost." [Leave] [Stay]

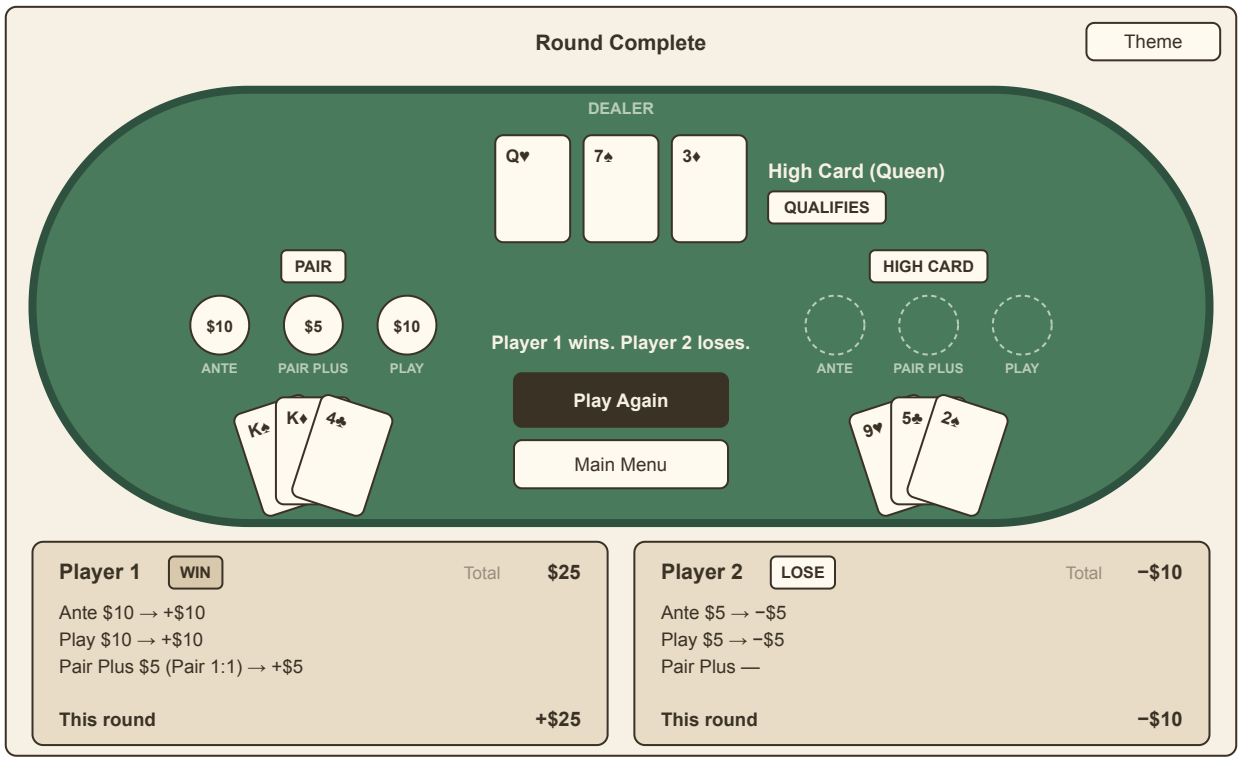
Screen 2 — Game Play (Step 2: Play or Fold)



Animation: cards slide from the dealer to each seat (TranslateTransition).

Animation: once both players decide, dealer cards flip face up (ScaleTransition).

Screen 3 — Results / Round Summary



Animation: winning chips and the Total briefly scale up (ScaleTransition).

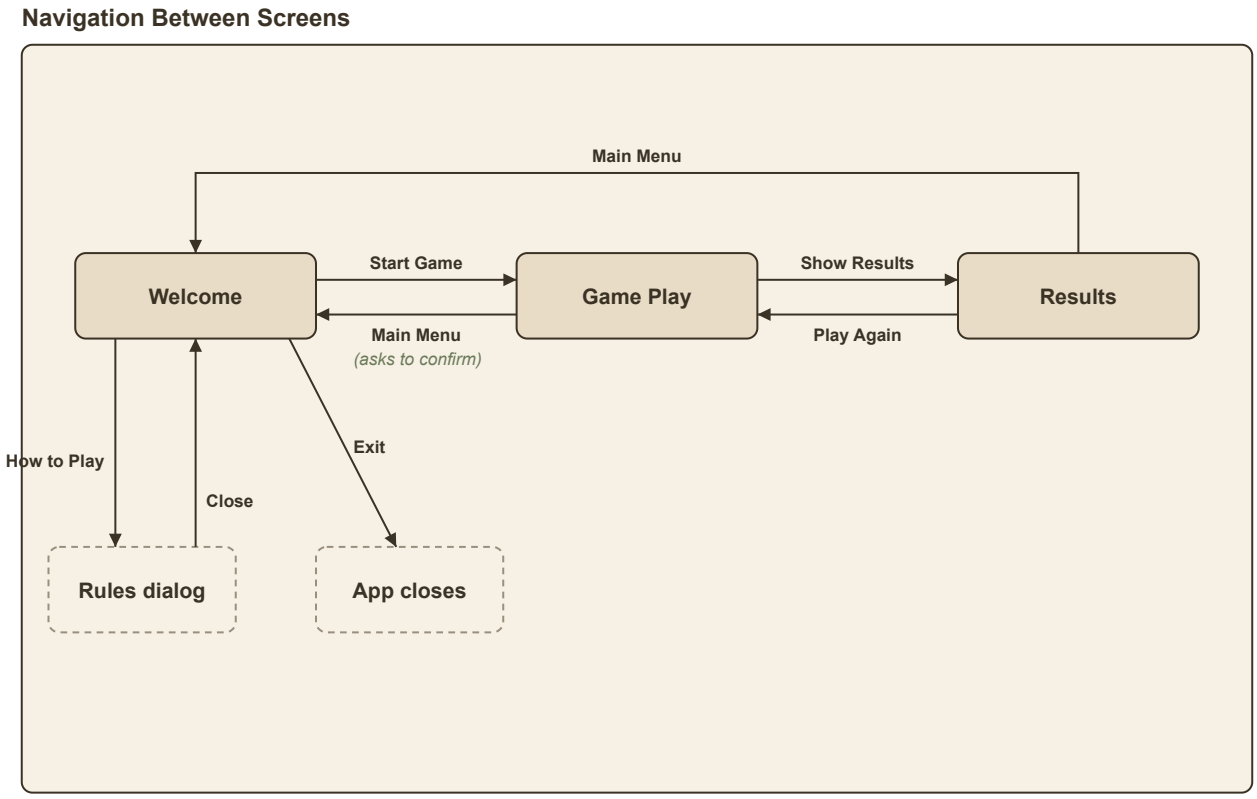
If the dealer does not qualify: "DOES NOT QUALIFY · Play returned, Ante carries over".

2. Alternate State

Alternate State — Player 1 Folded



3. Navigation



4. Software Architecture

- **Model:** `Card` (suit, value 2–14), `Deck` (52 cards; initialize, shuffle, reset), `Dealer` (owns the deck, deals 3-card hands), `Player` (hand, ante, pair plus, play wager, winnings).
- **Game logic and payouts:** `ThreeCardLogic`, with static `evalHand`, `evalPPWinnings` and `compareHands`.
- **Game service:** `GameService` holds the dealer, both players and the current phase. It takes bets, deals, records Play/Fold, resolves the round, updates winnings and starts the next round.
- **FXML controllers:** `WelcomeController`, `GameController`, `ResultsController`, one per screen. They handle events, call `GameService` and update the UI. They contain no rules or payout math.
- **Screen navigation:** `App` owns the `Stage`. `show(fxml)` swaps scenes and applies the current theme stylesheet.

Game rules stay independent of the UI because the model, logic and service classes do not use JavaFX. Controllers only call `GameService`, so the logic can be unit-tested without launching the UI.

5. Usability Testing

Tasks: (1) Start a new game. (2) Place Ante and Pair Plus bets. (3) Play or fold. (4) Find the round results. (5) Start another round, then return to the main menu.

> **TODO:** fill in from sessions with two students who are not on your team.

Participant 1:

Task	Completed?	Hesitation / confusion	Questions asked
1			
2			
3			
4			
5			

Suggestions:

Participant 2:

Task	Completed?	Hesitation / confusion	Questions asked
1			
2			
3			
4			
5			

Suggestions:

6. Usability Findings

1.
2.
3.

7. Wireframe Revisions

Revision 1: Before / After / Why:

Revision 2: Before / After / Why:

8. UI/UX Principles

- **Feedback:** the progress bar, the message on the felt and the player status labels always show the phase and the next action.
- **Error prevention:** bets are limited to $5–$25, invalid actions are disabled, and leaving mid-round asks for confirmation.
- **Consistency:** both seats share one layout, and the main action is always in the center of the table.
- **Visual hierarchy:** each screen has one filled primary button, and results lead with the outcome.
- **Accessibility:** controls are keyboard-accessible, and WIN/LOSE/FOLDED are written in text, not shown by color alone.